Graduate Algebra 1: midsem exam

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October 2026

1 Polynomials and field extensions

Claim 1. *If $F \subseteq K$ is a field extension of degree 2, then it is normal*

Proof. We have $K = F(\alpha)$. Let $p \in F[x]$ be the minimal polynomial of α. Then K is the splitting field of p and hence is normal. □

Claim 2. *Let $q(x) = x^4 - 2x^2 - 1$*

1. *$q(x)$ is irreducible over $\mathbb{Q}$*
2. *$q(x)$ does not split completely over $\mathbb{R}$*

Proof. Consider the following:

1. We have $q(x+1) = x^4 + 4x^3 + 4x^2 - 2$. Now use Eisenstein's criterion with $p = 2$ to show that $q(x+1)$ is irreducible. Since the map $p(x) \to p(x+1)$ is an automorphism of the ring $Q[x]$, $q(x)$ is also irreducible.
2. This is easy to see

□

2 Fields of prime characteristic

Claim 3. *Suppose F is finite and $a, b, c \in F^\times$. Consider the polynomials $p_1(x, y) = x^2 + y^2 - a$ and $p_2(x, y) = x^2 + by^2 - c$ in $F[x, y]$. Then in F, p_1 and p_2 have roots*

Proof. Consider the following:

- If F has characteristic 2, by the surjectivity of the Frobenius map, we can find solutions of the form $(\alpha, 0)$ to both equations
- Otherwise, use the fact that $|\{x^2 : x \in F\}| = \frac{q+1}{2}$

□

Claim 4. *Suppose F is a field of characteristic p and K is an algebraic closure. Let $f(x) = x^p - x - a \in F[x]$. Then we have:*

1. *$f(x) = (x+\alpha)(x+\alpha+1)\dots(x+\alpha+p-1)$ in $K[x]$, for some $\alpha \in K$*
2. *If $\alpha \notin F$ then f is irreducible over F*

Proof. Consider the following:

1. We have $f(x+1) = f(x)$
2. If $(x+\alpha+c_1)\dots(x+\alpha+c_m) \in F[x]$, then $m\alpha + c_1 + \dots + c_m \in F$. This forces $m = p$

□

Claim 5. *Suppose we have the following:*

- *p is odd, $K_1 = \mathbb{F}_p(t)$ and K_2 is an algebraic closure of K_1*
- *$f(x) = x^{2p} - t \in K_1[x]$*
- *$f(\theta) = 0$ with $\theta \in K_2$*

Then we have the following:

1. *f is irreducible in $K_1[x]$*
2. *$f(x) = (x-\theta)^p(x+\theta)^p$ in $K_2[x]$*
3. *θ is not purely inseparable over K_1*

Proof. Consider the following:

1. f is irreducible by Eisenstein's criterion
2. Use the Frobenius homomorphism
3. If $\theta^{p^n} = \theta^p t^{(p^{n-1}-1)} \in K_1$ then $\theta^p \in K_1$. But θ has degree $2p$ over K_1.

□

3 Modules

Suppose we have the following:

- k is a field
- $I = t^2k[t]$
- $R = k + I$

- Regard $k[t]$ as an R module using the inclusion $R \subseteq k[t]$

Claim 6. *The quotient vector space I/I^2 has basis $(t^2 + I^2, t^3 + I^2)$*

Claim 7. *I is an ideal but not a principal ideal in R*

Proof. Consider the following:

- It is easy to see that I is an ideal of R
- Suppose I is generated by a monic polynomial $p(t) \in I$. Then $p(t)$ is the lowest degree monic polynomial in I. Hence $p(t) = t^2$. But the element $t^3 \in I$ does not belong to t^2R

□

Claim 8. *$k[t]$ is not a free R-module*

Proof. Consider the following:

- The equation $(t^2p_2)p_1 - (t^2p_1)p_2 = 0$ shows that no two elements of $k[t]$ are linearly independent
- Suppose the module homomorphism $R \to k[t]$ given by $q(t) \to q(t)p(t)$ is surjective for some $p(t) \in k[t]$. We then get a surjective module homomorphism from R/I to $k[t]/I$. Since R/I is a 1 dimensional k vector space we cannot have such a k linear map

□

Claim 9. *There does not exist an R-module homomorphism $T : k[t] \to R$ such that $T(r) = r$ for all $r \in R$*

Proof. If T is such a map, we have $T(t^3) = t^3 = t^2T(t)$ which means that $T(t) = t \notin R$ which is a contradiction □